

Minimum central circles: chain order, seam obstructions, and exact finite certification

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Abstract

We minimize the radius of a central circle externally tangent to pairwise non-overlapping circles of radii $1, 2, \dots, n$. The adjacent-chain relaxation has an angular kernel with strict anti-Monge structure. Supnick’s classical theorem therefore supplies a fixed minimizing cyclic order at every central radius, and a monotonicity argument transfers this order to the chain-radius problem. The full geometry additionally requires every pairwise angular constraint. We give a self-contained criterion for feasibility of the canonical Supnick necklace on $\{k, \dots, n\}$ at its chain root, expressed by one seam deficit. For every fixed k the seam eventually fails and its failure persists. We prove the complete classifications for $k = 1, 2, 3$, whose first failures occur at $n = 8, 13, 17$, respectively. For the global problem, we certify rational brackets $L_n < R^*(n) \leq U_n$ of exact width 10^{-11} for every $3 \leq n \leq 14$. The lower proof combines independently enclosed angles, integer dynamic programming, and complete cyclic-order coverage; the upper proof uses exact rational Cartesian witnesses. It does not depend on floating-point pruning in the numerical search. We distinguish these finite brackets and fixed-order theorems from numerical descriptions of floating circles and from unresolved claims about all optimal placements.

Keywords: circle packing; discrete geometry; computational geometry; traveling salesman problem; Monge matrices; certified computation.

Mathematics Subject Classification (2020): 52C26, 52C15, 05C85, 90C27.

1 Introduction and contributions

Consider $n \geq 3$ circles with radii $1, 2, \dots, n$, each externally tangent to a central circle. Their interiors must be pairwise disjoint, and the central radius is to be minimized. Denote its global infimum by $R^*(n)$. The freedom to permute the surrounding circles makes this both a geometric feasibility problem and a discrete optimization problem. In particular, an order minimizing the angle needed by consecutive circles need not admit a non-overlapping placement at its chain-closing radius.

The problem and a conjectured “pyramid” order were posed by Dan in January 2023 on Mathematics Stack Exchange (question 4619480). The present paper gives a finite, standalone treatment. Its main certified numerical result is an enclosure of the global optimum, not an assertion that a printed decimal is an exact optimum:

$$L_n < R^*(n) \leq U_n, \quad U_n - L_n = \frac{1}{10^{11}}, \quad n = 3, \dots, 14. \quad (1)$$

All endpoints are stated explicitly in Section 7.

There are four mathematical contributions. First, the geometric angular kernel is strictly anti-Monge for every central radius. Applying the classical extreme-tour theorem of Supnick [1] and then comparing closure roots solves the adjacent-chain relaxation, including for arbitrary distinct positive radii. Second, we formulate full fixed-order feasibility using both directed angular constraints for every pair. Third, a single seam deficit characterizes feasibility at the chain root of the canonical order on $\{k, \dots, n\}$. Its eventual persistent failure is proved for each fixed k ; the complete strict classifications proved here are precisely those for $k = 1, 2, 3$. Fourth, an exact finite certificate proves (1) by combining exhaustive lower-bound coverage with rational geometric existence witnesses.

The new geometric conclusions are not part of Supnick’s tour theorem. That theorem supplies a minimizing order for a cost matrix with the specified structure. The identification of that structure in the angular kernel, the transfer from fixed radius to closure radius, the failure of the chain relaxation as full geometry, and the subsequent exact global certificates are separate steps. In particular, neither the chain theorem nor the seam theorem establishes floating behavior at global optima.

The Monge sign conventions and well-solvable traveling salesman problems are surveyed in [2]. The all-pairs formulation is a system of difference constraints, as in temporal constraint networks [3]. Related circle-packing work prescribes tangency structure. Mathews and Zymaris [4] generalize Descartes’ circle theorem to n -flowers, whose petals are tangent to the central circle and to their cyclic neighbors. Collins and Stephenson [5] compute circle packings realizing a specified pattern of tangencies. In Ringmin, the outer radii are fixed, the cyclic order must be chosen, and full pairwise non-overlap must be imposed without assuming tangency between consecutive outer circles. Container packing has different boundary conditions: here every surrounding circle must touch the same central circle, and angular order is a primary variable. We use no general packing theorem to infer global optimality.

An earlier finite preprint is available as arXiv:2607.28654v2. The separate asymptotic study, *Minimum central circles: an effective characterization of the global asymptotic constant*, is available as arXiv:2609.13630. No asymptotic theorem or conjectural asymptotic coefficient is needed for any result proved here.

2 Geometric angular reformulation

Place the central center at the origin. An outer circle of radius $a > 0$ has center $(R + a)(\cos \phi_a, \sin \phi_a)$. For $R, a, b > 0$ put

$$\theta_R(a, b) = 2 \arcsin \sqrt{\frac{ab}{(R + a)(R + b)}} \in (0, \pi). \quad (2)$$

Lemma 2.1 (Both directed arcs). *If the counterclockwise angular separation of two outer centers is $A \in [0, 2\pi]$, their circles have disjoint interiors if and only if*

$$\theta_R(a, b) \leq A \leq 2\pi - \theta_R(a, b). \quad (3)$$

The kernel is symmetric, strictly decreasing in R , and strictly increasing in each outer radius.

Proof. The squared center distance is $(a - b)^2 + 4(R + a)(R + b) \sin^2(A/2)$. Comparison with $(a + b)^2$ is equivalent to $\cos A \leq 1 - 2ab/((R + a)(R + b))$, which proves (3) on the full directed interval. The square-root argument in (2) lies in $(0, 1)$, and its monotonicities prove the rest. $\square$

For a cyclic order $\sigma = (s_0, \dots, s_{N-1})$ of at least three positive radii define

$$C_\sigma(R) = \sum_{i=0}^{N-1} \theta_R(s_i, s_{i+1}), \quad s_N = s_0.$$

It decreases continuously from $N\pi > 2\pi$ to zero. Thus $C_\sigma(R_{\text{chain}}(\sigma)) = 2\pi$ defines a unique positive *chain radius*. Let $R_{\text{full}}(\sigma)$ be the infimum of radii admitting an all-pairs-feasible placement in this order. Summing adjacent gap constraints, including the closing gap, gives

$$R_{\text{chain}}(\sigma) \leq R_{\text{full}}(\sigma), \quad \min_{\sigma} R_{\text{chain}}(\sigma) \leq R^*(n) = \min_{\sigma} R_{\text{full}}(\sigma). \quad (4)$$

For sufficiently large R , equal angular gaps give a feasible placement, since every $\theta_R(a, b)$ tends to zero. These infima are finite; none of the certificate arguments below presumes attainment of an infimum.

3 Anti-Monge structure and the Supnick chain order

Our sign convention calls a symmetric kernel anti-Monge when

$$\theta_R(a, b) + \theta_R(a', b') > \theta_R(a, b') + \theta_R(a', b) \quad (a < a', b < b'). \quad (5)$$

Lemma 3.1. *The kernel (2) satisfies (5) for every $R > 0$.*

Proof. Direct differentiation gives

$$\frac{\partial^2 \theta_R}{\partial a \partial b} = \frac{\sqrt{R}}{2\sqrt{ab}(R + a + b)^{3/2}} > 0. \quad (6)$$

Integrating over $[a, a'] \times [b, b']$ proves the strict inequality. Analytic diagonal values are allowed in this calculation; they are not self-pair constraints in the geometric model. $\square$

Here is an unambiguous definition of the relevant tour on N ranked vertices. Set $H = \lceil N/2 \rceil$. For successive $j = 0, 1, \dots$, append $1 + 2j$ to list A when it is at most H , and then append $N - 1 - 2j$ when it is greater than H . Construct list B by appending $2 + 2j$ when it is at most H , followed by $N - 2 - 2j$ when it is greater than H . Stop each list when neither entry qualifies. Concatenate A , the reverse of B , and N , and close the cycle. Every rank occurs once. Denote this rank cycle by σ_N^* . For example, $\sigma_4^* = (1, 3, 2, 4)$ and $\sigma_7^* = (1, 6, 3, 4, 5, 2, 7)$.

For a symmetric Monge cost matrix, Supnick's extreme-tour theorem identifies this rank cycle as a maximum-cost tour; see the maximum-tour discussion in [2]. Apply it to the Monge matrix $-\theta_R$. The sign reversal makes σ_N^* a minimum-angle tour. The classical tour theorem is the only imported ordering theorem.

Theorem 3.2 (Chain optimization). *For arbitrary distinct radii $0 < r_1 < \dots < r_N$, the order obtained by replacing rank i in σ_N^* by r_i minimizes $C_\sigma(R)$ at every $R > 0$, and hence minimizes $R_{\text{chain}}(\sigma)$. Its chain radius is a lower bound for the global geometric optimum. If its cumulative adjacent-angle placement is all-pairs feasible at that radius, the bound is attained and the placement is globally optimal.*

Proof. The fixed-radius assertion follows from (5) and Supnick's theorem. At $R = R_{\text{chain}}(\sigma)$ for any competing order, the minimizing tour has angle sum at most 2π . Strict decrease of its sum in R places its root no higher than $R_{\text{chain}}(\sigma)$. Now use (4). An all-pairs-feasible placement at the lower bound proves equality. $\square$

4 Fixed-order geometric feasibility

Fix R and σ , and unroll angles as $\phi_0 < \dots < \phi_{N-1} < \phi_0 + 2\pi$. For every $i < j$ impose

$$\theta_R(s_i, s_j) \leq \phi_j - \phi_i \leq 2\pi - \theta_R(s_i, s_j). \quad (7)$$

The positivity and upper bound already enforce the indicated order and wrap interval. By Lemma 2.1, this system is equivalent to full geometric feasibility. Written as $x_v - x_u \leq c_{uv}$ it is feasible exactly when its directed constraint graph has no negative cycle. Necessity follows by summing around a cycle; for sufficiency, distances from an added zero-edge source provide feasible potentials. Floyd–Warshall therefore gives a fixed-radius decision procedure using $O(N^3)$ arithmetic operations on the real edge costs. Numerical implementations still need their own error control. The constraint set expands with R , justifying bisection when suitably enclosed comparisons are available.

At the chain root the sum of all adjacent lower bounds is exactly 2π . Every feasible adjacent gap must therefore equal its lower bound. Thus failure of one nonadjacent constraint at the cumulative-angle placement cannot be repaired by redistributing gaps at that same root and order. This is the mechanism of the seam obstruction.

5 The seam criterion and the first three onsets

For integers $k \geq 1$ and $n \geq k+2$, let $\sigma_{k,n}$ be the rank cycle just defined, with rank i replaced by $k+i-1$. Write $R_{k,n} = R_{\text{chain}}(\sigma_{k,n})$ and define its seam deficit

$$\Delta_{k,n} = \theta_{R_{k,n}}(n, k) + \theta_{R_{k,n}}(k, n-1) - \theta_{R_{k,n}}(n, n-1). \quad (8)$$

The neighbors of k are $n-1, n$ in both parities.

Theorem 5.1 (Fixed-order seam criterion and persistence). *For every $k \geq 1$, $n \geq k+2$, a placement in order $\sigma_{k,n}$ is feasible at $R_{k,n}$ if and only if $\Delta_{k,n} \geq 0$. In that case the cumulative-angle placement is feasible and $R_{\text{full}}(\sigma_{k,n}) = R_{k,n}$.*

For every fixed k there is a finite $s_k \geq 4k+1$ such that the deficit is negative for all $n \geq s_k$. For $k+2 \leq n < s_k$ it is positive, except possibly for equality at $n = s_k - 1$. For $k = 1, 2, 3$ there is no integer equality case, and the full strict classifications are given in Table 1.

Table 1: Complete chain-root classifications for the three specified values of k . These are fixed-order results, not floating-circle thresholds.

k	$\Delta_{k,n} > 0$	$\Delta_{k,n} < 0$	s_k
1	$3 \leq n \leq 7$	$n \geq 8$	8
2	$4 \leq n \leq 12$	$n \geq 13$	13
3	$5 \leq n \leq 16$	$n \geq 17$	17

Appendix A contains a complete proof, including both parity edge lists, the two angular directions, the closing gap, small cycles, equality cases and six rational endpoint bridges. The key inequality is that every triangle defect is at least $\delta_R = \theta_R(n, k) + \theta_R(k, n-1) - \theta_R(n, n-1)$. A fan decomposition transfers this lower bound to every directed path. At the root, the two complementary paths have total length 2π , which proves sufficiency; the forced adjacent gaps prove necessity.

Persistence follows by comparing the strictly increasing chain root with a strictly decreasing algebraic threshold. This comparison, rather than monotonicity of the raw deficit, proves eventual failure. We give no explicit formula for s_k when $k \geq 4$ and do not exclude an equality case there. The criterion itself includes equality.

Corollary 5.2. *For $3 \leq n \leq 7$, $R^*(n) = R_{1,n}$ and the complete canonical necklace is a global optimum. For every $n \geq 8$, its cumulative-angle placement at $R_{1,n}$ is infeasible.*

Proof. Combine Theorems 3.2 and 5.1. The second assertion concerns the specified chain root, not a global optimizing contact graph. $\square$

6 Exact finite global certification

The finite proof has two independent mathematical obligations: exclude every order at and slightly above L_n , and exhibit a feasible placement at U_n . It uses the exact certificate data and verifier pinned in Section 10. Historical floating-point candidate generation, Stage-A pruning and retained Top-K lists are not premises of either obligation.

6.1 Necessary induced-cycle inequalities

In any feasible placement, take any subset with at least three members in its induced cyclic order. Its directed gaps sum to 2π , and each gap is at least the angle required for its endpoints. Consequently

$$\sum_{(a,b) \text{ consecutive in the induced cycle}} \theta_R(a,b) \leq 2\pi. \quad (9)$$

This necessity holds even if a directed gap exceeds π . The verifier tests the full order and the induced orders after deleting 1 or $\{1,2\}$, omitting subsets of size less than three. One strict violation suffices to exclude a full order. We do not assume these tests are sufficient for feasibility.

6.2 Independent outward enclosures

Let $S = 2^{128}$. For each unordered pair at both endpoints the certificate supplies integer bounds ℓ, u for $S\theta_R(a,b)$, with $0 < \ell \leq u < 3S$ and $u - \ell \leq 2$. Set $q = ab/((R+a)(R+b))$, computed as a rational number. The verifier proves

$$\cos(\ell/S) > 1 - 2q > \cos(u/S). \quad (10)$$

Since $\cos$ is strictly decreasing on $(0, 3)$, this places the true angle strictly inside the proposed interval. No generator accuracy is assumed.

Cosine is enclosed by its Taylor series using outward integer rounding on the internal grid $B = 2^{224}$. Starting with the exact term B , multiply successive absolute terms by $x^2/((2j+1)(2j+2))$, rounding the lower bound down and the upper bound up. Signed addition uses the opposite bound on a subtracted term. After 112 terms, the remaining terms decrease for $0 \leq x \leq 4$; the first omitted term bounds the alternating remainder with its proper sign. This argument does not require the early terms to decrease. Appendix B gives the recurrence explicitly.

The circumference bounds T_-, T_+ satisfy $6S < T_- < T_+ < 7S$. Opposite strict cosine signs at $T_-/(4S)$ and $T_+/(4S)$ enclose the unique zero $\pi/2$ in $(1.5, 1.75)$, proving $T_-/S < 2\pi < T_+/S$. All 908 pair intervals and this circumference interval are independently checked. Use $w_{ab} = \ell_{ab}$ at L_n as symmetric integer lower edge weights. A cycle cost strictly greater than T_+ proves a strict violation of (9).

6.3 Complete canonical coverage and exact pruning

For $n \leq 9$, pin the largest label first and retain one of each pair of reflected permutations. Distinct labels give exactly $(n-1)!/2$ classes. Every class receives an explicit induced-cycle violation.

For $n \geq 10$, delete radius 1. A full cyclic order becomes a skeleton on $\{2, \dots, n\}$; insertion into each of its $n-1$ gaps recovers all full orders, including insertion in the closing gap. Deletion is the inverse. Working modulo rotation and reflection preserves this bijection between full classes and canonical skeletons with a chosen gap. With at least three distinct skeleton labels, reflection fixes no oriented cycle.

Put $V = \{2, \dots, n-1\}$ and anchor the skeleton at n . Compute the forward dynamic program

$$\begin{aligned} D[\{j\}, j] &= w_{nj}, \\ D[M, j] &= \min_{i \in M \setminus \{j\}} (D[M \setminus \{j\}, i] + w_{ij}). \end{aligned} \quad (11)$$

Induction on $|M|$ shows that this is the minimum weight of a path from n through M ending at j . The root cycle cost is $\min_j (D[V, j] + w_{jn})$. For a prefix of cost c ending at j with unvisited set M , symmetry and path reversal give the exact minimum completion cost $D[M \cup \{j\}, j]$. Prune precisely when

$$c + D[M \cup \{j\}, j] > T_+. \quad (12)$$

Every true angular completion then violates the skeleton's necessary cycle inequality. Equality is retained. At the root use the root cycle cost. At a leaf the completion is the closing edge. All remaining choices are visited without a candidate cap. The DP has $O(n2^n)$ states and $O(n2^{2n})$ arithmetic operations; traversal can still be factorial.

A pruned prefix with h unvisited labels covers exactly $h!$ oriented completions. If their total is P and K canonical skeletons remain, the independently recomputed partition satisfies

$$P + 2K = (n-2)!, \quad \frac{P(n-1)}{2} + K(n-1) = \frac{(n-1)!}{2}. \quad (13)$$

The verifier also checks that P is even, there are exactly K reversed leaves, and exactly $K(n-1)$ explicit insertions with distinct normalized full-order classes. Every insertion must exhibit a strict induced-cycle violation. The complete traversal and valid pruning justify coverage; the factorial identity alone would not prove it. Trace hashes identify recomputed DP, pruning and witness streams, rather than replacing them.

6.4 Strictness for the infimum

Infeasibility at $L = L_n$ alone would not establish $L < R^*(n)$. Here there is a smallest positive integer margin m among all explicit violations and pruning bounds. Every covered order has a necessary cycle whose true angular excess at L is at least m/S . For $R \geq L$, differentiating (2) gives the rational bound

$$\begin{aligned} |\theta'_R(a, b)| &= \sqrt{\frac{q_R}{1-q_R}} \left(\frac{1}{R+a} + \frac{1}{R+b} \right) \\ &\leq d_{ab} := \max \left(1, \frac{q_L}{1-q_L} \right) \left(\frac{1}{L+a} + \frac{1}{L+b} \right). \end{aligned}$$

We used decreasing q_R and $\sqrt{z} \leq \max(1, z)$ for $z \geq 0$. Let $D_n = n \max_{a < b} d_{ab}$ and $\varepsilon_n = m/(2SD_n) > 0$. Each relevant cycle has at most n edges, so its violation persists on $[L, L + \varepsilon_n]$ with at least half the margin. At smaller positive radii the angles are larger. Therefore $R^*(n) \geq L + \varepsilon_n > L$. The verifier constructs this positive rational buffer and checks $\varepsilon_n < U_n - L_n$. No assertion about attainment is needed.

6.5 Exact rational existence witnesses

Each row contains at least one rational stereographic parameter t_r for every radius r , defining a center

$$C_r = (U_n + r) \left(\frac{1 - t_r^2}{1 + t_r^2}, \frac{2t_r}{1 + t_r^2} \right). \quad (14)$$

Exact rational arithmetic checks $\|C_r\|^2 = (U_n + r)^2$ and, for every $a < b$, $\|C_a - C_b\|^2 > (a + b)^2$. These centers alone prove existence at U_n . Half-plane and cross-product comparisons check the claimed polar order and distinct directions. Missing per-row witnesses are rejected; checking an empty list would not prove existence.

The data additionally contain scaled angular positions. All directed and wrap inequalities are checked with positive padding $\lceil S/10^{16} \rceil$, using upper pair-angle bounds and the lower circumference bound. This is constructive positive slack, not a negative error allowance. The angular and rational Cartesian witnesses need not describe exactly the same angles. There are 47 Cartesian witnesses in total. Their feasibility at U_n does not identify the minimizing orders or their contact graphs at $R^*(n)$.

7 Certified results for $3 \leq n \leq 14$

Theorem 7.1 (Computer-certified finite brackets). *For each integer $n = 3, \dots, 14$, the rational endpoints in Table 2 satisfy (1).*

Computer-assisted proof. The complete exact verifier checks the outward intervals, recomputes the lower coverage in Section 6, and validates at least one rational Cartesian witness per row. The strict-margin argument in Section 6.4 proves the open lower bound; the witnesses prove the closed upper bound. All twelve endpoint differences equal $1/10^{11}$ as rational numbers. The pinned data, code, and complete execution command are specified in Section 10. $\square$

Table 2: Exact global brackets. Every decimal is a rational endpoint; the certified statement is $L_n < R^*(n) \leq U_n$. The last column counts supplied feasible upper witnesses, not exact optimal orders.

n	L_n	U_n	Witnesses
3	0.26086956521	0.26086956522	1
4	0.84445358956	0.84445358957	1
5	1.69549408120	1.69549408121	1
6	2.79491951889	2.79491951890	1
7	4.15318955374	4.15318955375	1
8	5.76779428458	5.76779428459	1
9	7.72672655261	7.72672655262	1
10	9.97990738586	9.97990738587	4
11	12.48872048718	12.48872048719	6
12	15.25887043044	15.25887043045	9
13	18.31756304721	18.31756304722	10
14	21.66539518221	21.66539518222	11

Table 3 records the coverage sizes. For $n = 10, 11, 12$ the minimum skeleton cycle already exceeds the upper circumference bound, so the root DP comparison excludes every order without explicit insertions. For $n = 13, 14$ some skeletons survive and all their insertions are

tested. Across the twelve cases, 268,648 full classes are explicitly checked and 3,374,988,556 are covered in total. These are finite combinatorial counts, not evidence for any larger value of n .

Table 3: Complete lower coverage recomputed by the exact verifier. An en dash indicates direct full-order enumeration. Zero explicit orders at $n = 10, 11, 12$ means the root skeleton bound excludes the whole domain.

n	Full classes covered	Explicit full classes	Retained skeletons
3	1	1	—
4	3	3	—
5	12	12	—
6	60	60	—
7	360	360	—
8	2,520	2,520	—
9	20,160	20,160	—
10	181,440	0	0
11	1,814,400	0	0
12	19,958,400	0	0
13	239,500,800	2,952	246
14	3,113,510,400	242,580	18,660

The certificate does not assert exact decimal equality, uniqueness, or a complete list of optimizing orders. Corollary 5.2 gives an additional analytic characterization for $n \leq 7$; it is logically separate from the uniform bracket certificate.

8 Interpreting finite floating regimes

At a specified placement, call circle c *strictly floating* if every outer pair involving c is strictly separated; its required tangency to the central circle remains. An existential statement at the optimum would assert that there exists an optimal placement with this property. A universal statement would assert it for every optimal placement. Neither follows from a nonessential constraint, a numerical slack, or a witness at a radius above the optimum.

The public finite preprint and its saved numerical reconstructions describe the pattern in Table 4. We retain it as interpretation of those reconstructions. The floating entries for $n \geq 8$ are numerical descriptions, not additional exact consequences of Theorem 7.1. For $3 \leq n \leq 7$, the displayed canonical optimum has every adjacent pair tangent, so it has no strictly floating member; this is a statement about that exhibited optimum.

For an adjacent tangent pair a, b , the radius ρ at which insertion uses exactly their angular gap satisfies $\theta_R(a, \rho) + \theta_R(\rho, b) = \theta_R(a, b)$. Appendix A derives the corresponding pocket curvature by tangent addition, without importing a packing formula. Strict inequality in the direction $\theta_R(a, \rho) + \theta_R(\rho, b) < \theta_R(a, b)$ permits slack against the two boundary circles. Strict floating in the whole placement still requires all the other pair constraints.

The new upper witnesses illustrate the quantifier issue sharply: every outer pair is strictly separated at U_n , so every outer circle is strictly floating there under the placement-level definition. This does not imply that every circle can float at $R^*(n)$; slack can disappear as the radius decreases. Likewise, the exact seam onsets 8, 13, 17 are failures of three specified chain necklaces and are not thresholds at which particular circles must float in globally optimal configurations.

Table 4: Finite geometric interpretation of the historical reconstructed solutions (public finite preprint, arXiv:2607.28654v2). “Paid” and “free” describe comparison with a reduced-chain value. Except for the exhibited necklace in the first row, these are numerical descriptions.

n	Reported floating set	Interpretation of the reconstruction
3–7	$\emptyset$	Complete canonical necklace; analytic global optimum.
8, 9	$\{1\}$	Reduced necklace adjusted at a radius above its chain bound (paid placement).
10–12	$\{1\}$	Circle 1 accommodated at the reduced canonical chain value to reported numerical precision (free placement).
13	$\{1\}$	Second seam obstruction in the canonical order beginning at 2; a constrained reconstruction is used.
14	$\{1, 2\}$	Reconstruction with two reported floating circles and a constrained remaining necklace.

9 Limitations and open questions

The exact global certificate stops at $n = 14$. Extending its finite scope requires new complete lower coverage and new feasible upper witnesses. Numerical searches may guide those choices, but a best-known radius is only an upper bound when feasibility is established. The present certificate does not classify all optimizing orders, exact optimal contact graphs, or the dimension of the set of optimal placements.

The most direct geometric question is to prove or refute the existential floating descriptions at the exact optimum, with simultaneous strict slack against every other outer circle. Universal floating assertions require a separate argument. Fixed- k persistent seam failure does not prove an indefinite global floating cascade or repeated paid-then-free regimes.

The anti-Monge chain theorem extends to arbitrary distinct positive radii. The integer-family seam onset statements do not automatically extend to arbitrary sequences or deletions. Nor does any finite table establish an asymptotic coefficient. The separate asymptotic manuscript mentioned in the introduction treats that different question; no part of its proof is imported into the present finite certification.

10 Reproducibility statement

The complete computational inputs and exact verifier are available in the pinned, content-addressed Git commit `falker47/ringmin`, commit `6c16af4`. The full commit identifier is

`6c16af422d1cb38641c62d43b6e0e547921b9ba9`

From its repository root, Python 3.11 or later with only the standard library suffices for the complete command:

```
python -I -S verify_global_brackets.py
```

There is no partial-case or skip-frontier success mode in this entrypoint. It checks all twelve cases and their pinned input binding. The local reproduction for this manuscript used Python 3.14.3 and returned `PASS_GLOBAL_BRACKETS`, 908 angle intervals, 47 witnesses, 540 central tangencies, 3,004 outer pairs and 6,008 angular inequalities. No runtime claim is needed for the theorem.

The data file is

```
reproducibility/global_brackets/originals/source/
ringmin_global_interval_candidate.json
```

The SHA-256 of the canonical serialization of its certificate field is

6f6c15e1db037a3faadadc52893a9a90ae7b5
9d6d5b42e934c81c1a0abd21cc0

This identifies the certificate field, not the bytes of the whole file. The commit’s manifest identifies the preserved original files separately.

The verifier is independent of the production solver and original generator: it imports neither, uses cosine enclosures rather than the generator’s arcsine/Machin evaluations, and recomputes the lower dynamic program in the forward direction. The integrated verifier derives from an independent reviewer’s implementation; rerunning it is reproduction of that lineage, not a new independent design. The generator is parsed only as static data for input binding. Historical runtime labels, platform statements and execution authenticity are not certified by source hashes.

Regression tests at the same snapshot reject false brackets, missing witnesses, altered interval endpoints, collapsed directions, malformed coverage and trace metadata, and changed pinned inputs. Separate small enumeration oracles exercise the DP and closing-gap coverage. The manuscript package supplies its own build instructions and a source-to-claim map. This is an exact-arithmetic computer-assisted proof, not a proof-assistant formalization; finite tests and successful reproduction do not remove the need to review the mathematics and verifier.

Statements and Declarations

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Author contribution. Maurizio Falconi is the sole author and is responsible for the research and the submitted manuscript.

Data and code availability. All inputs required for the finite certificate and its verifier are available at the pinned, content-addressed Git commit in Section 10; this is not an archival repository deposit. The proof of the selected seam theorem is included in Appendix A.

Use of artificial intelligence. OpenAI ChatGPT/Codex assisted with mathematical exploration, software, computational experiments, manuscript drafting and repository organization, including preparation of this journal version. This assistance extended beyond copy editing. The author retains responsibility for the final text, proofs and code. Automated checks and AI-assisted review do not constitute independent journal peer review.

A Self-contained seam theorem and proof

A.1 Definitions and statement

All surrounding circles are externally tangent to a central circle of radius $R > 0$. For surrounding radii $a, b > 0$, define

$$\theta_R(a, b) = 2 \arcsin \sqrt{\frac{ab}{(R+a)(R+b)}} \in (0, \pi). \quad (\text{A.1})$$

If their centers have counterclockwise angular separation $A \in [0, 2\pi]$, the squared distance between them is $(R+a)^2 + (R+b)^2 - 2(R+a)(R+b)\cos A$. Thus their interiors are disjoint if and only if

$$\theta_R(a, b) \leq A \leq 2\pi - \theta_R(a, b). \quad (\text{A.2})$$

Indeed, the distance requirement is equivalent to $\cos A \leq 1 - 2ab/((R+a)(R+b)) = \cos \theta_R(a, b)$. In particular, **both** directed arcs between a pair must have length at least its required angle. Tangencies, including nonadjacent tangencies, are allowed.

Fix integers $k \geq 1$, $n \geq k+2$, and put $N = n - k + 1$. The **canonical Supnick cycle** $\sigma_{k,n}$ is defined by the following rank rule, which specifies every entry, without an order ellipsis. Put $H = \lceil N/2 \rceil$. For successive integers $j \geq 0$:

- In a list A , append $1 + 2j$ if it is at most H , then append $N - 1 - 2j$ if it is greater than H .
- In a list B , append $2 + 2j$ if it is at most H , then append $N - 2 - 2j$ if it is greater than H .

Stop each list when neither entry can be appended. Concatenate A , the reverse of B , and the singleton N ; replace each rank i by $k+i-1$, and join the last entry to the first. Low ranks partition $1, \dots, H$ by parity, high ranks partition $H+1, \dots, N-1$, and the appended rank is N . Hence this is a cycle through each radius once. Rotations and reversal represent the same undirected cycle. Subscripts on cycle entries are read modulo N .

Let $E_{k,n}$ be its undirected edge set, and define

$$C_{k,n}(R) = \sum_{(a,b) \in E_{k,n}} \theta_R(a, b), \quad C_{k,n}(R_{k,n}) = 2\pi, \quad R_{k,n} = R_{\text{chain}}(\sigma_{k,n}). \quad (\text{A.3})$$

Existence and uniqueness of this positive root are proved below. At any radius write

$$\delta_R = \theta_R(n, k) + \theta_R(k, n-1) - \theta_R(n, n-1), \quad \Delta_{k,n} = \delta_{R_{k,n}}. \quad (\text{A.4})$$

A **fully feasible placement in this fixed order** consists of angles $\phi_0 < \dots < \phi_{N-1} < \phi_0 + 2\pi$, with centers $(R + s_i)(\cos \phi_i, \sin \phi_i)$, where $(s_0, \dots, s_{N-1}) = \sigma_{k,n}$, such that (A.2) holds for every pair. The **cumulative-angle placement at the chain root** has $\phi_0 = 0$ and successive gaps $\theta_{R_{k,n}}(s_i, s_{i+1})$, including the closing gap. Define $R_{\text{full}}(\sigma)$ as the infimum of radii admitting a fully feasible placement in order σ .

Theorem (fixed-order seam criterion and persistence). For every pair of integers $k \geq 1, n \geq k+2$:

1. The positive chain root exists uniquely. The following are equivalent: $\Delta_{k,n} \geq 0$; the cumulative-angle placement at $R_{k,n}$ is fully feasible; some fully feasible placement of this order exists at $R_{k,n}$. When these hold, $R_{\text{full}}(\sigma_{k,n}) = R_{k,n}$.
2. For each fixed k , there is a finite integer $s_k \geq 4k+1$ such that $\Delta_{k,n} < 0$ for every $n \geq s_k$. For $k+2 \leq n < s_k$ the deficit is positive, with at most one possible exception: $\Delta_{k,s_k-1} = 0$. Thus feasibility at the chain root holds exactly before s_k , and failure persists thereafter.

3. For $k = 1, 2, 3$, the complete classifications are:

$$\begin{array}{c|c|c} k & \Delta_{k,n} > 0 & \Delta_{k,n} < 0 \\ \hline 1 & 3 \leq n \leq 7 & n \geq 8 \\ 2 & 4 \leq n \leq 12 & n \geq 13 \\ 3 & 5 \leq n \leq 16 & n \geq 17 \end{array}$$

In particular $s_1 = 8, s_2 = 13, s_3 = 17$, with no integer equality case for these three values of k .

The proof does not assert the absence of equality for every $k \geq 4$, nor give an explicit formula for their onsets. It establishes the weak criterion in part 1 even when equality is not excluded.

A.2 Exact edges, both parities, and growth of the chain root

Put $L = k + n$. Reading the rank rule gives the following edge lists; indexed families are listed in increasing order of i :

$$\begin{aligned} N = 2h : & \quad (k, n), (k + h - 1, k + h), \\ & \quad (i, L - 1 - i), \quad k \leq i \leq k + h - 2, \\ & \quad (i, L + 1 - i), \quad k + 1 \leq i \leq k + h - 1; \\ N = 2h + 1 : & \quad (k, n), \\ & \quad (i, L - 1 - i), \quad k \leq i \leq k + h - 1, \\ & \quad (i, L + 1 - i), \quad k + 1 \leq i \leq k + h. \end{aligned} \tag{A.5}$$

Here $h \geq 2$ in the even case and $h \geq 1$ in the odd case. To check the formula directly, consecutive low/high ranks inside either list alternate endpoint sums N and $N + 2$. After translation these are $L - 1$ and $L + 1$. The closing edge is (k, n) . At the join of the two lists, an even-sized cycle has the additional central edge $(k + h - 1, k + h)$; an odd-sized cycle completes the two indexed families instead. Their lower endpoints run through exactly the ranges in (A.5). The counts are $2 + (h - 1) + (h - 1) = 2h$ and $1 + h + h = 2h + 1$. The endpoint ranges give distinct edges, so none is omitted or repeated.

For $N = 3$ the cycle is $(k, k + 1, k + 2)$. For every $N \geq 4$, the rank list starts with $(1, N - 1)$ and ends with N . Consequently the two neighbors of k are exactly $n - 1, n$ in both parities, including $N = 3$.

For fixed positive a, b , (A.1) is continuous and strictly decreasing in R , with limits π at zero and zero at infinity. Hence $C_{k,n}$ decreases continuously from $N\pi > 2\pi$ to zero. This proves the asserted existence and uniqueness.

Lemma A.1 (growth without an ordering-optimality premise). For fixed k , $R_{k,n+1} > R_{k,n}$, and $R_{k,n} \rightarrow \infty$.

Proof. The kernel strictly increases in either surrounding radius, since $a/(R + a)$ and $b/(R + b)$ do so separately in (A.1). Compare (A.5) at n and $n + 1$, so L increases by one.

- If $N = 2h + 1$, match the seam and each indexed edge (a, b) to $(a, b + 1)$. The new even list contains all these edges and the additional central edge $(k + h, k + h + 1)$.
- If $N = 2h$, match the seam and each indexed edge to $(a, b + 1)$, retaining the old central edge $(k + h - 1, k + h)$ unchanged. That retained edge is the last member of the new lower-sum family. The only remaining edge in the new odd list is $(k + h, k + h + 1)$.

Thus for every $R > 0$, $C_{k,n+1}(R) > C_{k,n}(R)$. Strict decrease in R implies strict growth of the roots. No deletion of a vertex and no optimality over other cycles is used.

All endpoints are at least k ; at the root, $2\pi \geq 2N \arcsin(k/(R_{k,n} + k))$. Inversion on the increasing branch, followed by $\sin x \leq x$, gives

$$R_{k,n} \geq k(\csc(\pi/N) - 1) \geq k(N/\pi - 1) \longrightarrow \infty \quad (k \text{ fixed}). \quad (\text{A.6})$$

This completes the proof. $\square$

A.3 The least triangle defect and the fan identity

Lemma A.2 (triangle defect, including equality). For any three distinct members a, b, c of $\{k, \dots, n\}$, at every $R > 0$,

$$\theta_R(a, b) + \theta_R(b, c) - \theta_R(a, c) \geq \delta_R. \quad (\text{A.7})$$

Equality holds exactly when $b = k$ and $\{a, c\} = \{n-1, n\}$.

Proof. Suppress R from the notation. Differentiating (A.1) on the positive domain gives

$$\theta_1(a, b) = \frac{\sqrt{Rb/a}}{(R+a)\sqrt{R+a+b}} > 0, \quad \theta_{12}(a, b) = \frac{\sqrt{R}}{2\sqrt{ab}(R+a+b)^{3/2}} > 0. \quad (\text{A.8})$$

Exchange endpoints so that $a < c$. Decreasing the middle argument b to k gives $\theta(a, b) + \theta(b, c) - \theta(a, c) \geq H(a, c)$, strictly if $b > k$, where $H(x, z) = \theta(x, k) + \theta(k, z) - \theta(x, z)$. Analytic values such as $\theta(k, k)$ are legitimate here; they are not self-pair constraints. For $x, z \geq k$,

$$H_1(x, z) = \theta_1(x, k) - \theta_1(x, z) \leq 0, \quad H_2(x, z) = \theta_2(k, z) - \theta_2(x, z) \leq 0.$$

The inequalities are strict respectively when $z > k$ and $x > k$. Since $a \leq n-1$, $c \leq n$, $c > k$, and $n-1 > k$, increasing first a to $n-1$, then c to n , proves $H(a, c) \geq H(n-1, n) = \delta_R$. More explicitly the difference is

$$\int_a^{n-1} \int_k^c \theta_{12}(x, y) dy dx + \int_c^n \int_k^{n-1} \theta_{12}(x, y) dx dy. \quad (\text{A.9})$$

It vanishes exactly when $a = n-1, c = n$. Together with the strict middle-argument comparison this proves the equality statement. This argument covers every ordering of the three distinct radii. $\square$

For a simple path $P = (v_0, \dots, v_m)$, define its angular slack as the sum of its m edge angles minus $\theta_R(v_0, v_m)$. For $m \geq 2$, cancellation gives the exact fan identity

$$S_R(P) = \sum_{j=1}^{m-1} [\theta_R(v_0, v_j) + \theta_R(v_j, v_{j+1}) - \theta_R(v_0, v_{j+1})] \geq (m-1)\delta_R. \quad (\text{A.10})$$

All triples are distinct, so Lemma A.2 applies term by term. For $m = 1$ the slack and the empty sum are both zero. If $\delta_R = 0$ and $m \geq 3$, at least two distinct vertices serve as middle arguments; they cannot both equal k . Thus the slack is strictly positive. For $m = 2$, equality in (A.10) occurs only at the seam through k , up to reversal. No triangle inequality has been assumed: (A.7) permits a negative lower bound.

A.4 Full feasibility, closing gap and small cycles

Work at $R = R_{k,n}$, and abbreviate $\Delta = \Delta_{k,n}$. For two distinct positions $i < j$, the two cyclic paths are

$$P_+ = (s_i, s_{i+1}, \dots, s_j), \quad P_- = (s_j, s_{j+1}, \dots, s_{N-1}, s_0, \dots, s_i).$$

Their edge counts are $d = j - i$ and $N - d$. Both are simple, even when one uses the closing edge. If $\Delta \geq 0$, applying (A.10) separately to both paths shows that each angular length is at least $\theta_R(s_i, s_j)$. Their sum is exactly 2π by (A.3). Hence (A.2) holds for every pair in the cumulative-angle placement. It is unnecessary to decide which direction is the shorter one.

For adjacent endpoints the one-edge slack is zero. Its complement has exact slack $2\pi - 2\theta_R(s_i, s_{i+1}) > 0$, since $\theta_R < \pi$. This checks the upper constraint as well as the lower one, for the closing pair just as for every other adjacent pair. Formula (A.2) then proves the claimed Cartesian non-overlap, and the center radii enforce tangency to the central circle.

Conversely, suppose some placement of this order is feasible at the chain root. Unroll its angles and let $g_i > 0$ be all N consecutive gaps, with $g_{N-1} = \phi_0 + 2\pi - \phi_{N-1}$. Adjacent feasibility gives $g_i \geq \theta_R(s_i, s_{i+1})$. Therefore

$$\sum_i [g_i - \theta_R(s_i, s_{i+1})] = 2\pi - C_{k,n}(R) = 0. \quad (\text{A.11})$$

Every summand is nonnegative, so every gap, including the closing gap, equals its adjacent angle. The two-edge path $(n, k, n-1)$, which exists by Appendix A.2, consequently has length $A = \theta_R(n, k) + \theta_R(k, n-1)$. Feasibility of its endpoints requires $A \geq \theta_R(n, n-1)$, namely $\Delta \geq 0$. If $\Delta < 0$, the complementary arc $2\pi - A$ also exceeds the upper bound $2\pi - \theta_R(n, n-1)$. There is no freedom to repair this failure by changing gaps at the same radius and order. This proves all three equivalences in part 1.

At any feasible radius r for this order, summing adjacent constraints gives $C_{k,n}(r) \leq 2\pi$, hence $r \geq R_{k,n}$. Feasibility at $R_{k,n}$ when $\Delta \geq 0$ proves the asserted equality of the full and chain radii.

For clarity the small cycles are explicit:

- For $N = 3$, $\sigma = (k, k+1, k+2)$. All pairs are adjacent; each two-edge complement has slack $2\pi - 2\theta_R(a, b) > 0$. In particular $\Delta = 2\pi - 2\theta_R(k+2, k+1) > 0$. There is no nonadjacent pair and no equality case.
- For $N = 4$, $\sigma = (k, k+2, k+1, k+3)$. The pair $(k, k+1)$ has the two paths through $k+2$ and through $k+3$; the pair $(k+2, k+3)$ has the two paths through k and through $k+1$. These are precisely the four nonadjacent directed constraints. Lemma A.2 bounds all four defects below by Δ , with equality only on the path through k . In particular the seam complement has two edges, not three. Adjacent complements have the positive slack already computed. Appendix A.5 also gives $\Delta > 0$, since $n = k+3 \leq 4k$.

For any $N \geq 4$, even a hypothetical $\Delta = 0$ is sufficient: only the two-edge seam can have zero slack among multi-edge paths, by (A.10) and its equality discussion. When $\Delta > 0$, the least nonadjacent directed-path slack is exactly Δ , attained only by that seam up to reversal. Longer paths have slack at least 2Δ ; the other two-edge paths have strictly larger defects by Lemma A.2.

A.5 Algebraic threshold and eventual persistence

We derive the threshold identity rather than invoke a circle-packing formula. With $x = 1/R$, $u = 1/a$, $v = 1/b$, (A.1) gives

$$\tan \frac{\theta_R(a, b)}{2} = \sqrt{\frac{ab}{R(R+a+b)}} = \frac{x}{\sqrt{x(u+v)+uv}}. \quad (\text{A.12})$$

Set $q = \sqrt{x(u+v)+uv} > 0$ and $w = x + u + v + 2q > 0$. The positive square-root identities

$$\sqrt{xu + (x+u)w} = x + u + q, \quad \sqrt{xv + (x+v)w} = x + v + q$$

follow by squaring. Also

$$(x + u + q)(x + v + q) - x^2 = q(2x + u + v + 2q) > 0.$$

Consequently the sum of the half-angles for pairs $(a, 1/w)$ and $(1/w, b)$ has tangent x/q . Each half-angle lies in $(0, \pi/2)$; the positive denominator means their sum also lies there. It therefore equals $\theta_R(a, b)/2$, with no branch ambiguity. We have proved

$$\theta_R(a, 1/w) + \theta_R(1/w, b) = \theta_R(a, b). \quad (\text{A.13})$$

For the seam endpoints put

$$\alpha_n = \frac{1}{n} + \frac{1}{n-1}, \quad \beta_n = \frac{1}{n(n-1)}, \quad P_n(x) = x + \alpha_n + 2\sqrt{\alpha_n x + \beta_n}. \quad (\text{A.14})$$

This is the Descartes pocket curvature, but (A.13) proves every identity needed here independently of that interpretation. Strict increase of the two angles with the inserted radius gives, for every $R > 0$,

$$\text{sign } \delta_R = \text{sign}(k - 1/P_n(1/R)) = \text{sign}(P_n(1/R) - 1/k). \quad (\text{A.15})$$

The function P_n is continuous and strictly increasing for $x \geq 0$, and tends to infinity. Its value at zero is $(1/\sqrt{n} + 1/\sqrt{n-1})^2$, strictly decreasing in n . For $k \geq 1$, strict decrease of $t \mapsto 1/\sqrt{t}$ gives

$$P_{4k}(0) = \left(\frac{1}{\sqrt{4k}} + \frac{1}{\sqrt{4k-1}} \right)^2 > \left(\frac{2}{\sqrt{4k}} \right)^2 = \frac{1}{k},$$

$$P_{4k+1}(0) = \left(\frac{1}{\sqrt{4k+1}} + \frac{1}{\sqrt{4k}} \right)^2 < \left(\frac{2}{\sqrt{4k}} \right)^2 = \frac{1}{k},$$

where squaring preserves the strict inequalities because all terms are positive. Thus $P_{4k}(0) > 1/k > P_{4k+1}(0)$, and $\delta_R > 0$ for every $R > 0$ when $k+2 \leq n \leq 4k$; a unique positive crossing $P_n(x) = 1/k$ exists exactly when $n \geq 4k+1$.

On this latter domain set $c = 1/k$ and $q_n = \sqrt{\alpha_n c + \beta_n}$. Squaring the crossing equation produces $c + \alpha_n \pm 2q_n$. The plus root is extraneous because the unsquared equation requires $c - \alpha_n - x \geq 0$, while the plus root gives $-2(\alpha_n + q_n) < 0$. Since the unique positive crossing exists, it must be the minus root

$$\kappa_{k,n} = \frac{1}{k} + \frac{1}{n} + \frac{1}{n-1} - 2\sqrt{\frac{2n+k-1}{kn(n-1)}} > 0, \quad T_{k,n} = \frac{1}{\kappa_{k,n}}. \quad (\text{A.16})$$

One can also check its branch directly: $P_n(0) < c$ gives $q_n^2 - \alpha_n^2 = \alpha_n(c - \alpha_n) + \beta_n > 0$, and $\alpha_n \kappa_{k,n} + \beta_n = (q_n - \alpha_n)^2$; its positive square root is $q_n - \alpha_n$. From (A.15),

$$\text{sign } \Delta_{k,n} = -\text{sign}(R_{k,n} - T_{k,n}) \quad (n \geq 4k + 1). \quad (\text{A.17})$$

For fixed $x \geq 0$, both α_n and β_n strictly decrease with n , so $P_{n+1}(x) < P_n(x)$. Evaluating at the crossing and using strict increase in x shows $\kappa_{k,n+1} > \kappa_{k,n}$, hence $T_{k,n+1} < T_{k,n}$. Equation (A.16) gives $T_{k,n} \rightarrow k$. Lemma A.1 now implies that $D_{k,n} = R_{k,n} - T_{k,n}$ strictly increases to infinity on $n \geq 4k + 1$.

Define s_k as the first integer where $D_{k,n} > 0$. It exists; the deficits are negative from then on. Before it they are positive except possibly at one integer with $D_{k,n} = 0$. Such an equality, if present, must immediately precede s_k by strict increase. The earlier no-threshold domain has strictly positive deficits. This proves part 2, including every quantifier, without assuming monotonicity of the raw angular deficits.

A.6 Six exact bridges and the three onset classifications

By (A.17) and strict growth of $D_{k,n}$, it suffices to establish

$$\begin{aligned} R_{1,7} &< 6 < T_{1,7}, & T_{1,8} &< 51/10 < R_{1,8}, \\ R_{2,12} &< 17 < T_{2,12}, & T_{2,13} &< 14 < R_{2,13}, \\ R_{3,16} &< 32 < T_{3,16}, & T_{3,17} &< 32 < R_{3,17}. \end{aligned} \quad (\text{A.18})$$

Here is all the rational arithmetic needed to verify these bridges.

A.6.1 Threshold sides, with signs checked before squaring

For each row let r be its separator, let $h = 1/k + \alpha_n - 1/r > 0$, and put $v = 4(\alpha_n/k + \beta_n) > 0$. Then $\kappa_{k,n} - 1/r = h - \sqrt{v}$. The margins in Table 5 are strictly positive rational numbers:

Table 5: Exact threshold comparisons; all displayed square margins are positive.

(k, n)	r	h	v	Positive square margin	Conclusion
(1, 7)	6	8/7	4/3	$v - h^2 = 4/147$	$r < T_{1,7}$
(1, 8)	51/10	3061/2856	8/7	$h^2 - v = 47737/8156736$	$T_{1,8} < r$
(2, 12)	17	1381/2244	25/66	$v - h^2 = 239/5035536$	$r < T_{2,12}$
(2, 13)	14	643/1092	9/26	$h^2 - v = 673/1192464$	$T_{2,13} < r$
(3, 16)	32	69/160	17/90	$v - h^2 = 671/230400$	$r < T_{3,16}$
(3, 17)	32	691/1632	3/17	$h^2 - v = 7465/2663424$	$T_{3,17} < r$

All six rows lie in the positive-threshold domain, so reciprocation is legitimate. No sign is inferred by squaring quantities of unknown sign.

A.6.2 Chain sides, including every closing and central edge

For each edge $e = (a, b)$ in the **list order of** (A.5) put $z_e^2 = ab/((r+a)(r+b))$. In Table 6, the displayed integer vector divided by d defines the entire vector (q_e) . An upper row certifies $q_e^2 - z_e^2 > 0$ for every edge; a lower row certifies $z_e^2 - q_e^2 > 0$. All entries are positive. These are finite rational comparisons, using (A.5) and the displayed formula for z_e^2 .

For example, the first edge of the first row is (1, 7), giving $(28/100)^2 - 1/13 = 12/8125 > 0$. An even row includes its central edge as the second entry. The exact minimum square margin in each row is, respectively,

$$\frac{43}{30000}, \quad \frac{4333}{26270000}, \quad \frac{32}{359375}, \quad \frac{3}{32500}, \quad \frac{23}{70000}, \quad \frac{1}{5625}. \quad (\text{A.19})$$

For upper rows we use two elementary estimates. For $0 < z < 1/2$,

Table 6: Rational edge bounds, in the order of (A.5).

(k, n)	r	Direction	d	Numerators of (q_e) , in order
(1, 7)	6	upper	100	28,27,34,37,37,41,43
(1, 8)	51/10	lower	1000	316,466,307,390,428,414,462,487
(2, 12)	17	upper	1000	209,204,236,257,270,276,250,274,291,301,306
(2, 13)	14	lower	1000	245,348,240,278,304,320,330,291,320,340,353,361
(3, 16)	32	upper	100	17,23,17,19,20,21,22,22,20,21,22,23,24,24
(3, 17)	32	lower	100	17,16,18,20,21,22,22,22,19,21,22,23,24,24,24

$$\arcsin z < \frac{z}{\sqrt{1-z^2}} < \frac{2z}{\sqrt{3}} < \frac{6}{5}z. \quad (\text{A.20})$$

The first comparison integrates the strictly increasing derivative; the last follows by squaring positive quantities. For $0 < u \leq 1/3$,

$$(1 + 3u^2/5)^2(1 - u^2) - 1 = \frac{u^2(5 - 21u^2 - 9u^4)}{25} > 0.$$

Indeed the parenthesis decreases as a function of u^2 and equals $23/9 > 0$ at $u^2 = 1/9$. Taking positive square roots and integrating proves

$$\arcsin z < z + z^3/5 \quad (0 < z \leq 1/3). \quad (\text{A.21})$$

For lower rows simply use $\arcsin z > z$ on $(0, 1)$. The standard strict bounds $3 < \pi < 22/7$ suffice: the lower bound follows from the inscribed regular hexagon, and the upper has the elementary witness

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx > 0,$$

obtained by polynomial division and $\int_0^1 dx/(1+x^2) = \pi/4$. Table 7 gives an exact rational bound for every closure comparison.

Table 7: Exact chain comparisons for the six endpoint bridges.

(k, n)	Bound on $C_{k,n}(r)/2 = \sum_e \arcsin z_e$	Strict comparison
(1, 7)	$< (6/5) \sum q_e = 741/250$, all $q_e < 1/2$	$< 3 < \pi$
(1, 8)	$> \sum q_e = 327/100$	$> 22/7 > \pi$
(2, 12)	$< \sum (q_e + q_e^3/5) = 1457520693/500000000$, all $q_e < 1/3$	$< 3 < \pi$
(2, 13)	$> \sum q_e = 373/100$	$> 22/7 > \pi$
(3, 16)	$< \sum (q_e + q_e^3/5) = 14885133/5000000$, all $q_e < 1/3$	$< 3 < \pi$
(3, 17)	$> \sum q_e = 63/20$	$> 22/7 > \pi$

Strict decrease of $C_{k,n}$ converts these to the root sides of (A.18). Appendices A.6.1 and A.6.2 therefore prove all six bridges without numerical approximations to roots or transcendental functions.

A.6.3 Completing the integer ranges

For $k = 1, 2, 3$, let s be respectively 8, 13, 17. The two bridges for that k give $D_{k,s-1} < 0 < D_{k,s}$. Strict increase of $D_{k,n}$ on $n \geq 4k + 1$ supplies the same strict signs throughout the respective earlier and later ranges. On $k + 2 \leq n \leq 4k$, Appendix A.5 already gives strict

positivity of Δ . Equation (A.17) and the equivalence in Appendix A.4 now prove the entire table in part 3. No integer lies between the two endpoints of each bridge, so no equality case remains for these three k .

B Outward integer cosine arithmetic

For completeness, the enclosure used in (10) can be implemented using the following exact recurrence. Write $x^2 = a/d$ with $a \geq 0, d > 0$, and let $B = 2^{224}$. The j th absolute series term is $t_j = x^{2j}/(2j)!$. Start $l_0 = u_0 = B$, and set

$$l_{j+1} = \left\lfloor \frac{al_j}{d(2j+1)(2j+2)} \right\rfloor, \quad u_{j+1} = \left\lceil \frac{au_j}{d(2j+1)(2j+2)} \right\rceil.$$

Induction gives $l_j \leq Bt_j \leq u_j$ for every j . Form lower and upper sums through $j = N - 1$ by using (l_j, u_j) for even j and $(-u_j, -l_j)$ for odd j . For $N = 112$ and $0 \leq x \leq 4$, the tail beginning with t_N decreases, since $t_{N+1}/t_N = x^2/((2N+1)(2N+2)) < 1$ and subsequent ratios are smaller. As N is even, the remainder lies in $[0, t_N]$: add u_N only to the upper sum. Division by B gives a rigorous interval for $\cos x$. For an odd cutoff one would instead subtract u_N from the lower sum. All rounding is integer floor or ceiling, so no library floating-point or transcendental error assumption enters this calculation.

For the actual circumference check the scaled endpoints are

$$\begin{aligned} T_- &= 2138057168129933495719360746323741566601, \\ T_+ &= 2138057168129933495719360746323741566602. \end{aligned}$$

The interval checks require strict inequalities; an inconclusive enclosure is a verification failure, not permission to infer a sign. Integer arithmetic also evaluates every pruning margin exactly. Rational arithmetic evaluates (14), the strict squared-distance comparisons and the buffers in Section 6.4. Thus analytic enclosure, combinatorial coverage and geometric existence are separate, explicitly checked parts of the proof.

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